CSE 219 — Signals and Linear Systems

Lec 8 · z-Transform: ROC, Inverse, Properties, LTI Systems

Lesson notes · Exam approach · TF practice with solutions · Quick notes (Ashraf Sir's portion)

How to Read These Notes

EXAM FOCUS	Exam importance, where it appears in the TF collection
MEMORIZE	Rules, tables and formulas to memorize
TRAP	Common mistakes and exam traps
EXAM-READY ANSWER	Model wording you can write in the exam
NOTE	Extra explanation / clarification
Grey box, monospace	Code, worked tables, derivations and hand-drawn sketches
Centered grey box	Formula to remember / final numerical result
Purple heading / ✓ green	Practice questions / their solutions

Contents

Exam Map — What the TF Papers Ask — z-transform questions and their weight

Lesson 17 — Definition, ROC and Standard Transforms — $z = e^{sT}$, $a^n u[n]$, left-sided, finite sequences, ROC properties 1–9

Lesson 18 — The Inverse z-Transform — contour formula, partial fractions, power series (log)

Lesson 19 — Properties and LTI Systems — ROC and causality/stability, pole-zero plots, difference equations

Quick Notes — z-Transform — merged revision notes + key diagrams

> **NOTE** Slide numbers are PDF page numbers of 8 - Z-Transform.pdf (47 pages, “Lec 8”); the printed slide number is one less. The deck mirrors the Laplace transform: right-sided ⇒ outside a circle, left-sided ⇒ inside, two-sided ⇒ an annulus; the unit circle plays the role of the jω-axis.

Exam Map — What the TF Papers Ask

Session (paper)	Questions answered by this note
2023-24 (CSE 219)	Q8(a) All possible ROCs of $X(z) = (1+2z^{-1}-z^{-2})/(1-(11/6)z^{-1}+z^{-2}-(1/6)z^{-3})$ (12) · Q8(b) Poles 1/3, 2/3, 2: ROC for right-sided, FT converges, FT diverges, left-sided (8) · Q8(c) z-transform of $\|n\|(½)^{\|n\|}$; pole-zero plot (10) · Q8(d) Laplace vs z-transform; why z for discrete time (5)
2022-23 (CSE 219)	Q1(a) z-transform and ROC of $x[n] = a^n u[n]$ (10) · Q3(a) $x[n] = a^n$, $0 \le n \le N-1$: show the ROC excludes z = 0 (10) · Q3(b) Derive the contour-integral formula of the inverse z-transform (10) · Q3(c) Inverse z-transform of $\log(1+az^{-1})$, $\|z\|>\|a\|$ (15)

EXAM FOCUS The z-transform appears only in the two newest CSE 219 papers, always in Ashraf's part: 2022-23 Q1(a), Q3 (35 marks) and 2023-24 Q8 (35 marks). Learn: $a^n u[n]$ with its ROC, the three ROCs of a rational X(z), partial fractions in z^{-1}, the unit-circle test for the DTFT, and the power-series method ($\log(1 + az^{-1})$).

Lesson 17 — Definition, ROC and Standard Transforms

Lec 8 sl. 4–26

z^n is the eigenfunction of discrete LTI systems: $y[n] = H(z)z^n$ with $H(z) = \Sigma h[k]z^{-k}$. Sampling e^{st} at $t = nT$ gives $(e^{sT})^n$, so $z = e^{sT} = e^{\sigma T}e^{j\omega T} = re^{j\theta}$ (sl. 6): the jω-axis maps to the unit circle $|z| = 1$, the left half-plane to the interior, a vertical strip to an annulus; s and $s + j2\pi/T$ give the same z, which is why z is used for sequences (sl. 7–8).

$$\mathbf{X(z) = \Sigma_{n=-\infty}^{\infty} x[n]z^{-n} \qquad X(re^{j\omega}) = F\{x[n]r^{-n}\} \quad (r = 1: \text{DTFT})}$$

The DTFT exists iff the ROC contains the unit circle $|z| = 1$

Sequence	X(z)	ROC
$a^n u[n]$	$1/(1 - az^{-1}) = z/(z - a)$	$\|z\| > \|a\|$ (pole a, zero 0; exterior of a circle)
$-a^n u[-n-1]$	$1/(1 - az^{-1})$	$\|z\| < \|a\|$ (same expression, interior)
$7(1/3)^n u - 6(1/2)^n u$	$z(z - 2/3)/((z - 1/3)(z - 1/2))$	$\|z\| > 1/2$
$(1/3)^n \sin(\pi n/4)u[n]$	$(\sqrt{2}/6)z^{-1}/(1 - (\sqrt{2}/3)z^{-1} + z^{-2}/9)$	$\|z\| > 1/3$
$\delta[n]$; $\delta[n-1]$; $\delta[n+1]$	1; z^{-1}; z	all z; all $z \neq 0$; all finite z
a^n, $0 \le n \le N-1$	$(1 - a^N z^{-N})/(1 - az^{-1}) = z(1 - a^N z^{-N})/(z - a)$	all $z \neq 0$; N−1 zeros on $\|z\| = a$ at $ae^{j2\pi k/N}$
$b^{\|n\|}$, $0 < b < 1$	$(1-b^2)/((1-bz^{-1})(1-bz))$	$b < \|z\| < 1/b$ (no ROC if $b > 1$)

MEMORIZE A z-transform = expression + ROC. Geometric series: $\Sigma(az^{-1})^n = 1/(1 - az^{-1})$ for $|az^{-1}| < 1$. Left-sided: $-\Sigma_{m\ge1}(a^{-1}z)^m = 1/(1 - az^{-1})$ for $|z| < |a|$.

ROC Properties (sl. 17–26)

#	Property
1	ROC is a ring centred at the origin (depends on $r = \|z\|$ only)
2	ROC contains no poles (poles bound it)
3	Finite-duration x ⇒ ROC = all z except possibly z = 0 (samples at $n > 0$ give z^{-n}) and/or ∞ (samples at $n < 0$). $N_1 \ge 0 \Rightarrow \infty$ included; $N_2 \le 0 \Rightarrow 0$ included
4	Right-sided ⇒ exterior of a circle $\|z\| > r_0$ (includes ∞ iff causal)
5	Left-sided ⇒ interior $0 < \|z\| < r_0$ (includes 0 iff anti-causal)
6	Two-sided ⇒ annulus $r_1 < \|z\| < r_2$ (or empty)
7–9	Rational: bounded by poles or reaching 0/∞; right-sided ⇒ outside the outermost pole; left-sided ⇒ inside the innermost non-zero pole

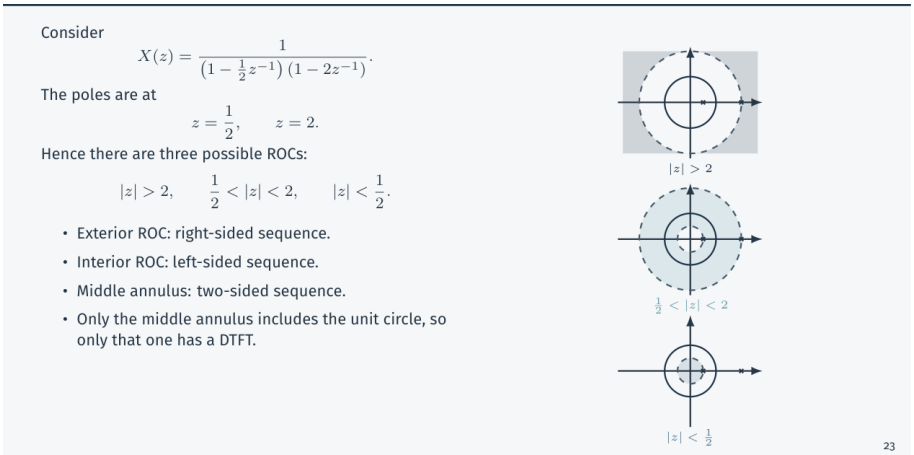


Figure 17.1 — Three ROCs of one X(z): exterior (right-sided), annulus (two-sided, has a DTFT), interior (left-sided) (Lec 8, sl. 26)

Exam Approach

1. Write the geometric series that defines X(z) and state the convergence condition — that is the ROC.
2. For a finite sequence: list the powers of z; positive powers (n < 0) exclude ∞, negative powers (n > 0) exclude 0.
3. For X(z) = N/D: poles = roots of D (in z, not in z^{-1}!); possible ROCs = regions separated by the pole circles.
4. State which ROC contains |z| = 1 (⇒ DTFT exists).

Practice — Lesson 17

(a) Actual — CSE 219 2022-23 Q1(a), Q3(a). (b) Actual — CSE 219 2023-24 Q8(a), Q8(d).

(a) (i) z-transform and ROC of $x[n] = a^n u[n]$. (ii) $x[n] = a^n$ for $0 \le n \le N-1$: show that the ROC of X(z) cannot include the origin. (10 + 10)

(b) Find all possible two-sided... sequences with $X(z) = (1 + 2z^{-1} - z^{-2})/(1 - (11/6)z^{-1} + z^{-2} - (1/6)z^{-3})$: for each, the ROC and x[n]. Differentiate the Laplace and z-transforms and say why z is preferred for discrete time. (12 + 5)

✓ Solution

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(a)(i) X(z) = Σ_{n=0}^{∞} aⁿz^{-n} = Σ(az⁻¹)ⁿ = 1/(1 − az⁻¹) = z/(z − a),   |az⁻¹| < 1 ⇔ |z| > |a|
      pole z = a, zero z = 0; ROC outside the circle through the pole; contains |z| = 1 iff |a| < 1.
(ii) X(z) = Σ_{n=0}^{N−1} aⁿz^{-n} = 1 + az⁻¹ + a²z⁻² + … + a^{N−1}z^{−(N−1)}.  For N ≥ 2 the term a z⁻¹ → ∞ as z → 0
     (|z|⁻¹ → ∞), so X(0) is not finite ⇒ z = 0 cannot belong to the ROC. The sum is finite for every z ≠ 0,
     so ROC = |z| > 0. (Closed form: X = z(1 − aᴺz⁻ᴺ)/(z − a): N−1 poles at the origin; the pole at z = a is
     cancelled by a zero.)
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(b) Denominator: z³ − (11/6)z² + z − 1/6 = (z − 1)(z − 1/2)(z − 1/3) ⇒ poles 1, 1/2, 1/3
    X = 6/(1 − z⁻¹) − 3/(1 − ½z⁻¹) − 2/(1 − ⅓z⁻¹)   [x[0] = lim X = 1 = 6 − 3 − 2 ✓]
    Four ROCs (rings between the pole circles); poles inside the ROC's inner side give right-
sided terms:
    (1) |z| > 1       : x[n] = [6 − 3(1/2)ⁿ − 2(1/3)ⁿ] u[n]
right-sided, causal
    (2) 1/2 < |z| < 1 : x[n] = −6u[−n−1] − 3(1/2)ⁿu[n] − 2(1/3)ⁿu[n]
two-sided
    (3) 1/3 < |z| < 1/2: x[n] = −6u[−n−1] + 3(1/2)ⁿu[−n−1] − 2(1/3)ⁿu[n]
two-sided
    (4) |z| < 1/3     : x[n] = −6u[−n−1] + 3(1/2)ⁿu[−n−1] + 2(1/3)ⁿu[−n−1]
left-sided
    Rule: a pole on the inner side of the ROC (|p| < |z|) → right-sided term Apⁿu[n];
          a pole on the outer side (|p| > |z|) → left-sided term −Apⁿu[−n−1].
    The pole at z = 1 lies on the unit circle, so |z| = 1 belongs to no valid ROC: the DTFT of
none of the four exists.
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(d) Laplace: continuous-time, $X(s) = \int x\, e^{-st}dt$, $s = \sigma + j\omega$, ROC = vertical strips/half-planes, stability ↔ ROC ∋ jω-axis, rational in s. z-transform: discrete-time, $X(z) = \Sigma x[n]z^{-n}$, $z = re^{j\omega}$, ROC = rings/exteriors/interiors, stability ↔ ROC ∋ unit circle, rational in z (or z^{-1}), related by $z = e^{sT}$. Why z for DT: the map $s \to e^{sT}$ is many-to-one (s and $s + j2\pi/T$ give the same sequence e^{snT}), so the s-plane would count the same discrete signal infinitely often; z removes that redundancy, turns difference equations into algebraic ones, uses the finite sum $\Sigma x[n]z^{-n}$ (no integrals) and makes the unit circle the frequency axis.

Lesson 18 — The Inverse z-Transform

Lec 8 sl. 28–32

1. Contour Formula (sl. 28)

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z = re^{jω} in the ROC: x[n]r^{-n} = (1/2π)∫_{2π} X(re^{jω})e^{jωn}dω  (inverse DTFT)
multiply by r^n:  x[n] = (1/2π)∫_{2π} X(re^{jω})(re^{jω})^n dω
z = re^{jω} ⇒ dz = jz dω ⇒ dω = dz/(jz):
x[n] = (1/2πj) ∮_C X(z) z^{n-1} dz     (C: counter-clockwise circle inside the ROC; evaluate by residues)
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2. Partial Fractions (sl. 29–31)

$X(z) = (3 - (5/2)z^{-1})/((1 - \frac{1}{4}z^{-1})(1 - \frac{1}{3}z^{-1})) = 1/(1 - \frac{1}{4}z^{-1}) + 2/(1 - \frac{1}{3}z^{-1})$. ROC $|z| > 1/3$ ⇒ both right-sided: $x = (\frac{1}{4})^n u[n] + 2(\frac{1}{3})^n u[n]$. ROC $\frac{1}{4} < |z| < \frac{1}{3}$: $x = (\frac{1}{4})^n u[n] - 2(\frac{1}{3})^n u[-n-1]$. ROC $|z| < \frac{1}{4}$: both left-sided: $-(\frac{1}{4})^n u[-n-1] - 2(\frac{1}{3})^n u[-n-1]$. The expansion never changes; the ROC decides each term.

3. Power Series (sl. 31–32)

The coefficient of z^{-n} is x[n] (positive powers of z are negative times). $4z^2 + 2 + 3z^{-1} \Rightarrow 4\delta[n+2] + 2\delta[n] + 3\delta[n-1]$. $1/(1 - az^{-1}) = 1 + az^{-1} + a^2z^{-2} + \ldots$ ($|z| > |a|$) ⇒ $a^n u[n]$; for $|z| < |a|$ expand in z/a ⇒ $-a^n u[-n-1]$.

```
X(z) = log(1 + az^{-1}), |z| > |a|:   log(1 + v) = Σ_{n≥1} (-1)^{n+1}v^n/n  (|v| < 1), v = az^{-1}
(|v| < 1 ⇔ |z| > |a|)
X(z) = Σ_{n≥1} (-1)^{n+1} (a^n/n) z^{-n}   ⇒   x[n] = (-1)^{n+1} a^n/n for n ≥ 1,  0 for n ≤ 0
i.e. x[n] = (-1)^{n+1}(a^n/n) u[n - 1]   (checked numerically: Σ ... = log(1 + a/|z|))
```

Exam Approach

1. Rational X: partial fractions in z^{-1} (write with unit constant terms), then use the ROC for each term.
2. Log/irrational X: expand in a power series of z^{-1} valid in the ROC ($|\ldots| < 1$) and read off the coefficients.
3. Check $x[0] = \lim_{z\to\infty} X(z)$ for causal sequences.

Practice — Lesson 18

(a) Actual — CSE 219 2022-23 Q3(b), Q3(c). (b) Practice set 11.1–11.5 (first two).

(a) Derive the contour-integral formula for the inverse z-transform; find the sequence for $X(z) = \log(1 + az^{-1})$, $|z| > |a|$. (10 + 15)

(b) Invert $(2 - z^{-1})/((1 - \frac{1}{4}z^{-1})(1 - \frac{3}{4}z^{-1}))$, $|z| > \frac{3}{4}$; and $1/((1 - 2z^{-1})(1 - \frac{1}{3}z^{-1}))$, $1/3 < |z| < 2$.

✓ Solution

(a) Derivation and the logarithm series in the boxes above (the log series needs $|az^{-1}| < 1$, which is exactly the ROC $|z| > |a|$).

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(b) (2 - z^{-1})/((1 - ¼z^{-1})(1 - ¾z^{-1})) = 1/(1 - ¼z^{-1}) + 1/(1 - ¾z^{-1})   [A = (2-4)/(1-3)=1, B = (2-4/3)/(1-1/3)=1]
    |z| > 3/4: x[n] = [(¼)^n + (¾)^n] u[n]
    1/((1-2z^{-1})(1-⅓z^{-1})) = (6/5)/(1 - 2z^{-1}) - (1/5)/(1 - ⅓z^{-1})   [A = 1/(1-1/6) = 6/5, B = 1/(1-6) = -1/5]
    1/3 < |z| < 2: the pole 2 is outside ⇒ left-sided; pole 1/3 inside ⇒ right-sided:
    x[n] = -(6/5)2^n u[-n-1] - (1/5)(⅓)^n u[n]
```

Lesson 19 — Properties and LTI Systems

Lec 8 sl. 34–47

Property	Sequence	Transform	ROC
Linearity	$ax_1 + bx_2$	$aX_1 + bX_2$	$\supseteq R_1 \cap R_2$
Time shift	$x[n - n_0]$	$z^{-n_0}X(z)$	R (except 0, ∞)
z-scaling	$a_0^n x[n]$	$X(z/a_0)$	$\lvert a_0 \rvert R$
Reversal	$x[-n]$	$X(z^{-1})$	R^{-1}
Conjugation	$x^*[n]$	$X^*(z)$	R
Time expansion	x[n/k] (zeros between)	$X(z^k)$	$R^{1/k}$
Convolution	$x_1 * x_2$	X_1X_2	$\supseteq R_1 \cap R_2$
First difference	$x[n] - x[n-1]$	$(1 - z^{-1})X$	R (+ 0 deleted, z = 1 may be added)
Accumulation	$\Sigma_{k \le n} x[k]$	$X/(1 - z^{-1})$	$\supseteq R \cap \{\lvert z \rvert > 1\}$
z-differentiation	$n\,x[n]$	$-z\,dX/dz$	R
Initial value	$x[n] = 0$ for $n < 0$	$x[0] = \lim_{z\to\infty} X(z)$	

Example 10.18: $a^n u[n] \leftrightarrow 1/(1 - az^{-1}) \Rightarrow n\,a^n\,u[n] \leftrightarrow -z\,d/dz[1/(1 - az^{-1})] = az^{-1}/(1 - az^{-1})^2$.

LTI Systems (sl. 43–47)

- $Y(z) = H(z)X(z)$; if the ROC contains $|z| = 1$, $H(e^{j\omega})$ is the frequency response.
- **Causal** ⇔ ROC is the exterior of a circle including ∞ (for rational H: outside the outermost pole, numerator degree ≤ denominator degree in z).
- **Stable** ⇔ ROC contains the unit circle. Causal + rational: stable ⇔ all poles inside $|z| = 1$.
- $H = 1/(1 - 2r\cos\theta\, z^{-1} + r^2z^{-2})$: poles $re^{\pm j\theta}$; causal ⇒ stable iff $r < 1$.
- Difference equation $\Sigma a_k y[n-k] = \Sigma b_k x[n-k] \Rightarrow H(z) = \Sigma b_k z^{-k}/\Sigma a_k z^{-k}$; the equation gives the expression, **not the ROC** (causality chooses it).
- Ex 10.25: $y[n] - \frac{1}{2}y[n-1] = x[n] + \frac{1}{3}x[n-1] \Rightarrow H = (1 + \frac{1}{3}z^{-1})/(1 - \frac{1}{2}z^{-1})$; step response ($|z| > \frac{1}{2}$): $Y = H/(1 - z^{-1}) = (8/3)/(1 - z^{-1}) - (5/3)/(1 - \frac{1}{2}z^{-1}) \Rightarrow y[n] = (8/3)u[n] - (5/3)(\frac{1}{2})^n u[n]$.

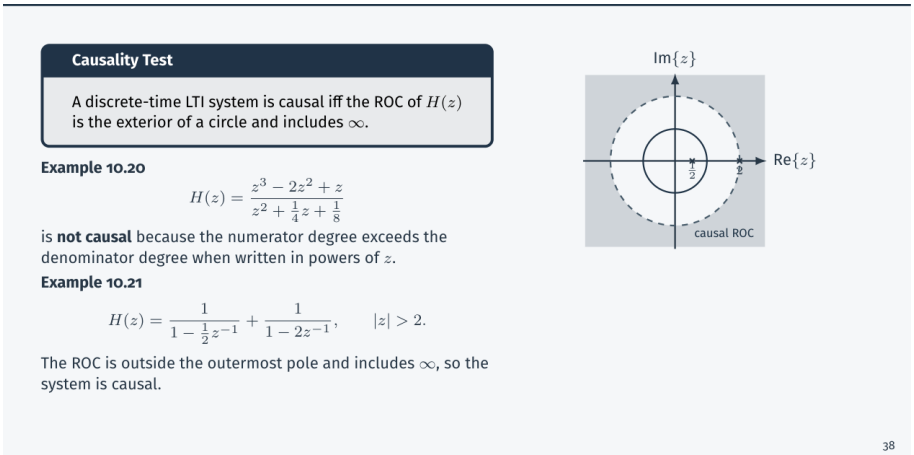


Figure 19.1 — Causality and stability from the ROC of H(z) (Lec 8, sl. 44)

TRAP • Poles are roots in z: $1 - (11/6)z^{-1} + z^{-2} - (1/6)z^{-3} = 0 \Leftrightarrow z^3 - (11/6)z^2 + z - 1/6 = 0$. • The ring that contains the unit circle is the only one with a DTFT. • ROC of a sum can exceed the intersection if poles cancel. • Left-sided/anti-causal sequences use $u[-n-1]$ with a minus sign.

Exam Approach

1. Identify the pole radii; list the rings.
2. Right-sided → outside the largest pole; left-sided → inside the smallest; DTFT exists ⇔ unit circle in the ROC.
3. For $|n|b^n$-type sequences split $n \geq 0$ and $n < 0$, transform each, intersect the ROCs (a ring).

Practice — Lesson 19

(a) Actual — CSE 219 2023-24 Q8(b), Q8(c). (b) Practice set 12.1, 12.4.

(a) (i) Fig. 2 shows the pole-zero plot of Y(z) (poles at 1/3, 2/3 and 2). What can be said about the ROC if: (i) y[n] is right-sided; (ii) the FT of y[n] converges; (iii) the FT does not converge; (iv) y[n] is left-sided? (8) (ii) z-transform and pole-zero plot of x[n] = $|n|(1/2)^{|n|}$. (10)

(b) H(z) = $(1 - 2z^{-1})/((1 - \frac{1}{2}z^{-1})(1 - 2z^{-1}))$: valid ROCs, causality, stability and DTFT; h[n] = $(2/3)^n u[n] - (4/3)^n u[-n-1]$: H(z), ROC, causal? stable? DTFT?

✓ Solution

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(a)(i) right-sided ⇒ ROC is the exterior of the outermost pole: |z| > 2
    (ii) FT converges ⇔ ROC contains |z| = 1 and no pole: 2/3 < |z| < 2
    (iii) FT does not converge ⇔ the ROC excludes |z| = 1: |z| > 2, or |z| < 1/3, or 1/3 < |z|
< 2/3
    (iv) left-sided ⇒ interior of the innermost pole: |z| < 1/3
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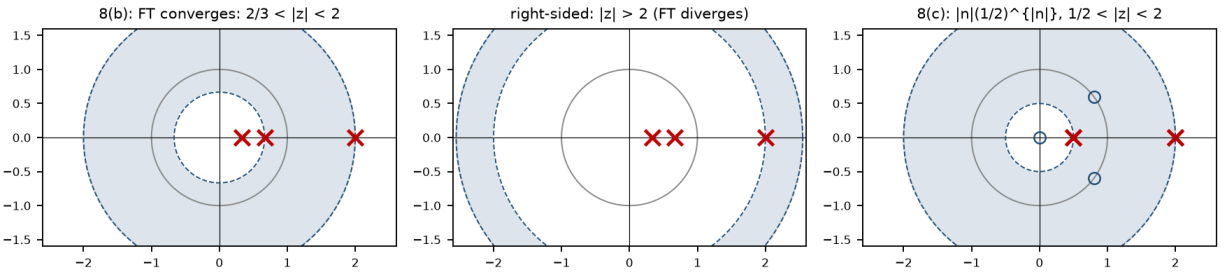


Figure 19.2 — Pole-zero plots with the ROC shaded: 8(b)(ii), 8(b)(i), and 8(c)

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(ii) x[n] = |n|aⁿ-type, a = 1/2:  for n ≥ 0: n(½)ⁿu[n] ↔ ½z⁻¹/(1 − ½z⁻¹)², |z| > ½
     for n < 0: x[n] = (−n)(½)^{−n} = (−n)2ⁿ = m(½)^m with m = −n ≥ 1:  Σ_{m≥1} m(½)^m z^{m} =
½z/(1 − ½z)², |z| < 2
     X(z) = ½z⁻¹/(1 − ½z⁻¹)² + ½z/(1 − ½z)² = 2z(5z² − 8z + 5)/((z − 2)²(2z − 1)²),   ½ < |z| <
2
     poles: z = ½ (double), z = 2 (double);  zeros: z = 0 and z = 0.8 ± 0.6j (on the unit
circle, 5z² − 8z + 5 = 0)
     the ROC is an annulus containing |z| = 1 ⇒ the DTFT exists. Check: X(1) = 4 =
Σ|n|(½)^{|n|} ✓
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(b) H = (1 − 2z⁻¹)/((1 − ½z⁻¹)(1 − 2z⁻¹)): the zero at z = 2 cancels the pole at z = 2 ⇒ H =
1/(1 − ½z⁻¹)
    cancelling: H = 1/(1 − ½z⁻¹) with ROC |z| > ½ (causal; contains the unit circle ⇒ stable,
DTFT exists) or |z| < ½
    (anti-causal, unstable). Without cancelling, the rings are |z| > 2, ½ < |z| < 2, |z| < ½
(the 2-pole counts as a boundary).
    h[n] = (2/3)ⁿu[n] − (4/3)ⁿu[−n−1]: H = 1/(1 − (2/3)z⁻¹) + 1/(1 − (4/3)z⁻¹), ROC 2/3 < |z| <
4/3 (the two ROCs
    overlap in a ring) ⇒ two-sided ⇒ non-causal; the ring contains |z| = 1 ⇒ stable and the
DTFT exists.
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Quick Notes — z-Transform

Revision points for Lessons 17–19.

L17 · ROC

- $X = \Sigma x[n]z^{-n}$; $z = e^{sT}$; DTFT ⇔ ROC ∋ $|z| = 1$.
- $a^n u \leftrightarrow 1/(1-az^{-1})$, $|z| > |a|$; $-a^n u[-n-1]$, $|z| < |a|$; right → exterior, left → interior, two-sided → ring.
- Finite sequence: all z except 0 (n > 0 terms) / ∞ (n < 0 terms).

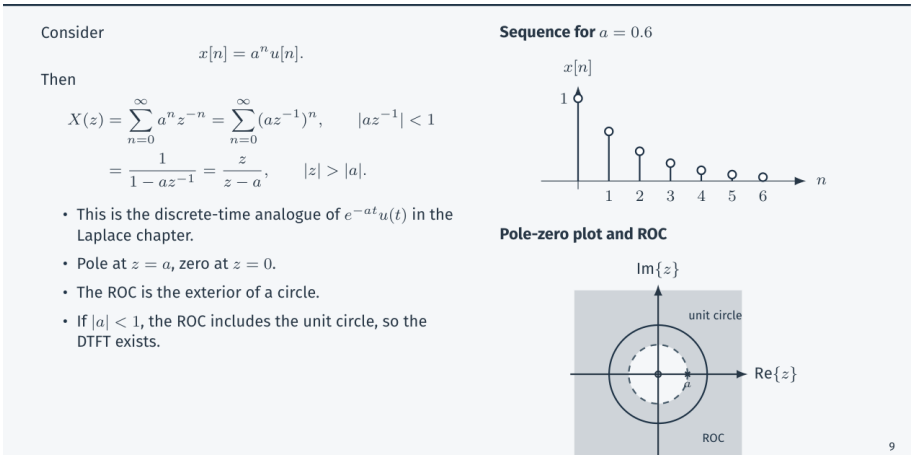
L18 · Inverse

- $x[n] = (1/2\pi j)\oint X z^{n-1}dz$; partial fractions in z^{-1}; pole inside the ROC's inner edge → right-sided term.
- $\log(1 + az^{-1})$, $|z| > |a| \Rightarrow (-1)^{n+1}a^n/n$, $n \geq 1$.

L19 · Systems

- Causal ⇔ exterior ROC incl. ∞; stable ⇔ ROC ∋ unit circle; causal + stable ⇔ poles inside $|z| = 1$.
- 8(a) 2023-24: poles 1, ½, ⅓: four ROCs; $X = 6/(1-z^{-1}) - 3/(1-\frac{1}{2}z^{-1}) - 2/(1-\frac{1}{3}z^{-1})$.
- 8(c): $X = 2z(5z^2-8z+5)/((z-2)^2(2z-1)^2)$, $\frac{1}{2} < |z| < 2$.

Key Diagrams to Reproduce



z-transform of $a^n u[n]$: pole-zero plot and ROC (Lec 8, sl. 12)